CLUB club

2026-11-05

2026-11-05 class

Today I'll give an introduction to the `dark-matter` scripts

WTF is dark-matter?

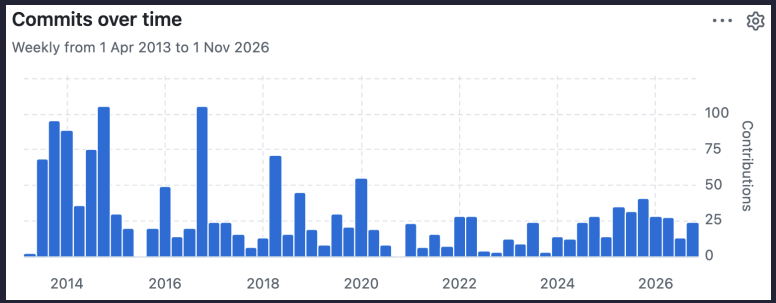
A suite of Python scripts (and library functions) for bioinformatics work.

You don't need to be a Python programmer to use the scripts. But you need Python installed.

Stats

`dark-matter` was written over the last 13 years (first commit 27.06.2013).

It provides 108 scripts, and has 87,267 lines of code and 2,627 tests.



Installing dark-matter

```
$ pixi add dark-matter
# Or 'pixi upgrade dark-matter' if you already have it installed.

$ pixi shell

$ dark-matter-version.py
7.5.8
```

You can find the Python source on GitHub at https://github.com/acorg/dark-matter

dark-matter is famously undocumented!

Or it was until now.

I got Claude to help with a basic categorization and description of all the scripts. So there is now a fairly minimal (but at least accurate) main README at https://acorg.github.io/dark-matter/

Click on the `Scripts` link on the left to see a listing of the available scripts. Each script (that supports `--help`) can be clicked on to see its `--help` output.

Or go to https://github.com/acorg/dark-matter/ and in the `doc` directory you'll see a link to `scripts.md`.

Today we'll look briefly at 15 scripts

Simple utilities

- aa-compare.py
- aa-info.py
- fasta-count-chars.py
- fasta-count.py
- fasta-ids.py
- fasta-lengths.py

More FASTA tools

- compare-sequences.py
- fasta-identity-table.py
- filter-fasta.py

Command runners

- make-consensus.py
- run-bowtie2.py

Tree tools

- curate-tree-ete.py
- tree-info.py

SAM/BAM files

- sam-coverage-depth.py
- sam-reference-read-counts.py

aa-compare.py

aa-info.py

Print information about amino acids, based on names or codons.

fasta-count-chars.py

Print a summary of the characters in the sequences in FASTA input.

fasta-count.py

Count the number of sequences in FASTA input.

fasta-ids.py

Print the sequence IDs from FASTA input.

fasta-lengths.py

Print the sequence IDs and lengths from FASTA input.

compare-sequences.py

fasta-identity-table.py

Make an HTML table showing the identity between sequences in FASTA input.

filter-fasta.py

Provides many ways to filter FASTA.

make-consensus.py

run-bowtie2.py

curate-tree-ete.py

tree-info.py

sam-coverage-depth.py

sam-reference-read-counts.py

Upcoming classes

- de novo assembly.
- Protein-level matching.